

Three Worlds of 2041

How AI could reshape our industry, our customers and our planet over the next 15 years

THREE SCENARIOS, NOT THREE PREDICTIONS

On 6 April 2026, four people watched Earth set behind the Moon. Fifteen years from now, the Moon may be a place where people go to work. This is not a dream. It is three futures we can already see coming, and the choices that decide which one we get.

IN THIS BRIEFING today's signals, our method, three worlds of 2041, the odds, and what to do from Monday.



Mohammad

Senior Futures Consultant, Em, Oli & Timmonds

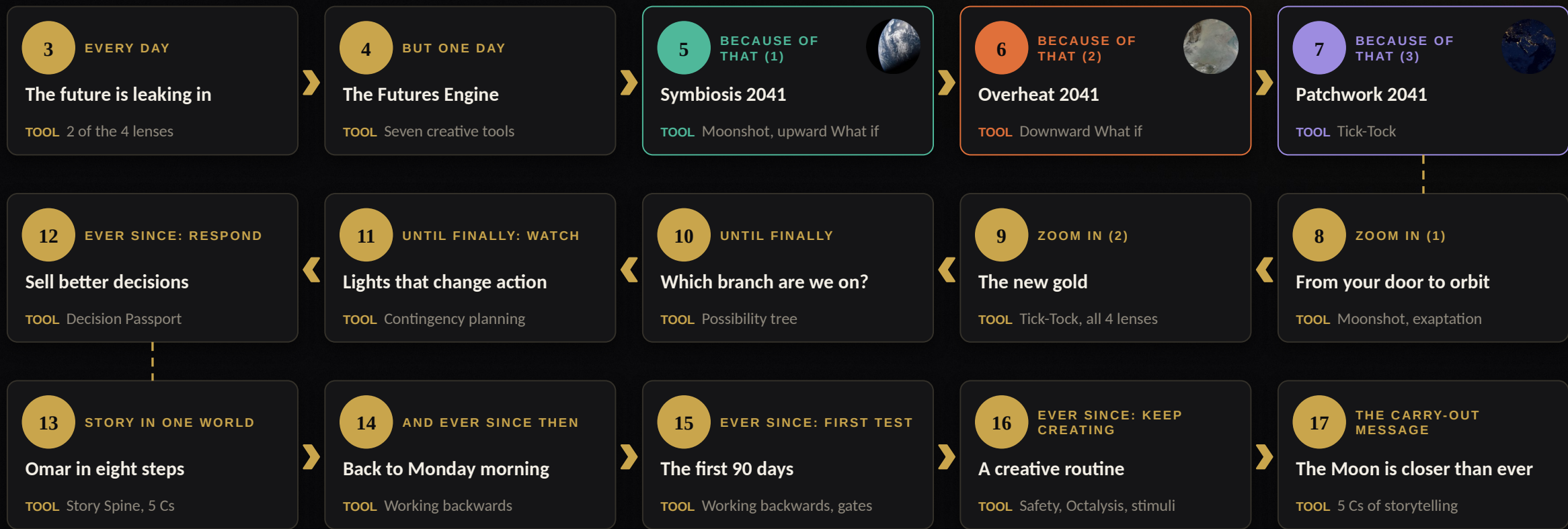
Part 1 | The Creative Solution

Earth sets behind the Moon, photographed by the Artemis II crew, 6 April 2026. Image: NASA

NEXT the story map: how this briefing unfolds in fifteen steps.

One story, fifteen steps

Our brief: imagine the world in 15 years' time, considering AI's impact on our industry and on the whole world, and decide what E,O&T should build now without betting on one forecast. The story uses an adapted, branching Story Spine: the three "Because of that" steps are alternative worlds. Facts and projections are tagged; the 2041 worlds are imagined scenarios.

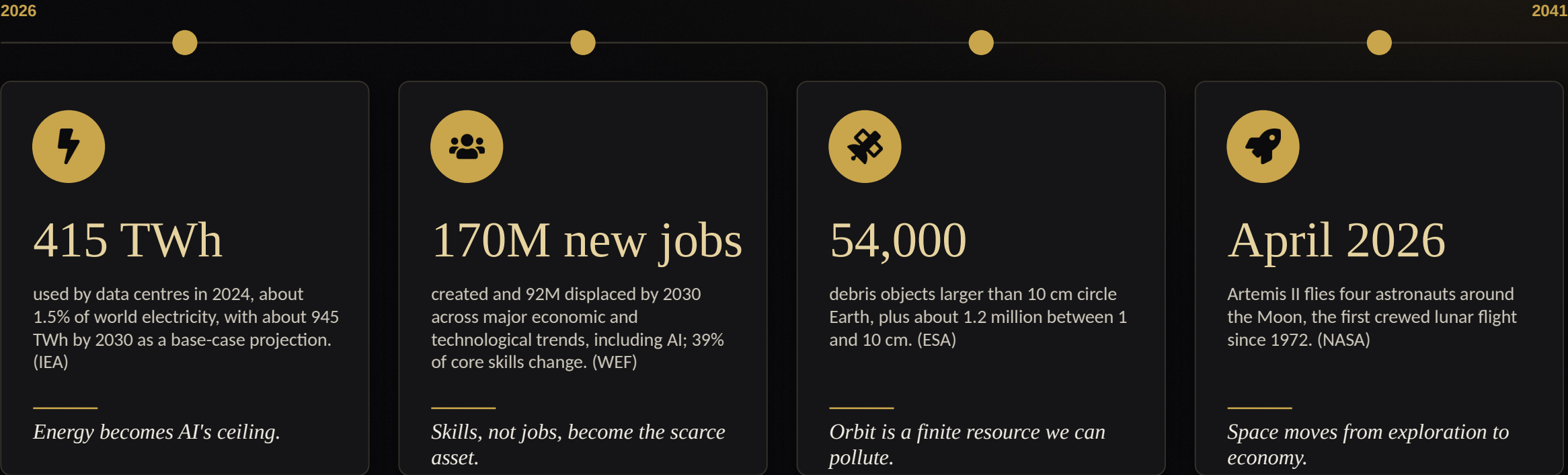


SCOPE Industry: strategic consulting | Clients: energy, logistics and digital services | World: work, daily life, infrastructure and the space frontier

NEXT we start with the facts: four signals already visible in 2026.

The future is already leaking in

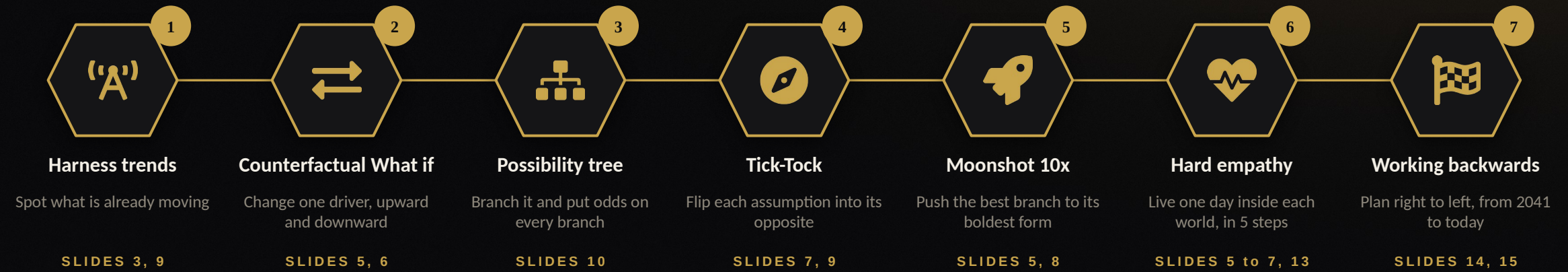
Four signals we can measure in 2026, found with two of the 4 lenses of innovation: harnessing trends and challenging orthodoxies. Every world grows out of them.



 **Challenging orthodoxies.** The old belief "AI is software" no longer holds. AI is an energy, materials and orbit industry, and that is exactly where our three worlds split apart.

The E,O&T Futures Engine

How the signals become worlds. Seven tools work like gears, and each produces a visible output on the slides shown. We first ran a controlled What if (keep AI access fixed, change only energy headroom), then combined both uncertainties into three worlds.



 **AI "bad copying"**

We asked AI for dozens of variants of each world and kept the strange ones. Its mistakes became raw material.

 **Exaptation**

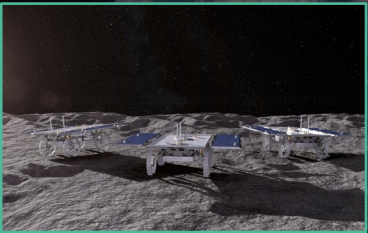
Borrowed from air traffic control (one network, many vehicles) and multiplayer game design (rules that shape behaviour).

SYMBIOSIS

2041

Moonshot thinking: the upward What if

Shared, verified AI with abundant clean energy



Robots on the Moon (NASA concept)



A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Layla, 34, grid engineer, Riyadh

Scene: 6 am. My AI agent has already sold my solar surplus and drafted my report.

Feel: free, and strangely idle.

Make sense: the machines now do the routine half of my job.

Worry: what am I still for?

Act: I join the council that designs our city's digital twin.

A future we want to be part of

The best case. We push the engine's upper branch to a moonshot, then live one day inside it through hard empathy.



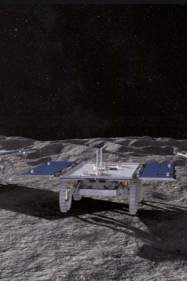
Mirror Cities

Customers try a flat or a lunar hotel in a live digital twin before buying. White-hat Octalysis: each purchase plants a real tree.



Life and transport

Electric pods and air taxis on one AI network, borrowed from air traffic control. Car parks become parks.



Space and the Moon

A crewed base at the lunar south pole mines ice for fuel. Robots build; the first tourists arrive.



Our industry

AI writes the analysis; consultants own the judgement. E,O&T move: launch a future-design practice.

5 Cs

Character

Layla, and every client like her

Conflict

AI's hunger for energy

Cure

Clean power, efficient chips

Change

Abundance, shared widely

Carry-out

Build the future we want

Still hard here: e-waste, attention overload, disputes over lunar mining.

OVERHEAT

2041

Counterfactual: the downward What if
Concentrated, opaque AI with scarce, polluting energy



Hardware in low orbit (NASA, ISS)



A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Omar, 41, logistics manager

Scene: in the depot, two delivery alerts contradict each other.
Feel: anxious; the rationing app also limits when I can charge the vans.
Make sense: one of the alerts may be synthetic.
Worry: I could send a driver onto an unsafe route.
Act: I verify both alerts through our data cooperative before dispatch.

Every utopia has a shadow

The honest shadow of World 1. Same engine, downward What if, told as a chain of cause and effect.

HOW IT HAPPENED, ONE "BECAUSE OF THAT" AT A TIME



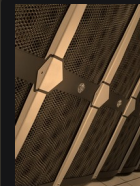
Customers
The metaverse is an escape; black-hat Octalysis (scarcity, loss) keeps people hooked.



Life and transport
Empty robotaxis gridlock the cities. The rich fly; everyone else waits.



Space and the Moon
Low orbit is a junkyard; the lunar base waits.



Our industry
Reports cost pennies. E,O&T move: verification and resilience services.

	Character	Conflict	Cure	Change	Carry-out
5 Cs	Omar	AI outgrows clean power	None, in time	Trust and orbit lost	Plan for the shadow first

PATCHWORK

2041

Tick-Tock: the middle path

Trust verified inside regional blocs, mixed energy between them

One planet, three rulebooks.



A DAY IN 2041 | HARD EMPATHY IN 5 STEPS

Sara, 29, product designer, Stockholm

Scene: my AI agent is doing my weekly shopping.

Feel: relaxed, until it refuses my favourite app.

Make sense: that app has no trust seal in our bloc.

Worry: my own product could be locked out the same way.

Act: I design one product that passes all three blocs' trust rules.

Neither heaven nor collapse

The middle path, built by flipping two of today's facts with Tick-Tock. Here trust, not technology, decides who wins.

TICK People buy from companies.

TOCK AI agents buy from AI agents, so firms must design for machine customers.

TICK Data is free.

TOCK Data is owned and paid for, like energy.



Customers

Regional metaverses joined by one digital identity, so trust becomes the product.



Life and transport

Driverless in some regions, manual in others; the best cities blend both.



Space and the Moon

Shared on paper; blocs compete quietly for lunar ice and landing sites.



Our industry

Winners translate between blocs. E,O&T move: a cross-bloc standards advisory.

5 Cs

Character

Sara

Conflict

One world, many rulebooks

Cure

Shared trust seals

Change

Identity is the product

Carry-out

The future is a negotiation

From your door to orbit

Borrow the delivery-network logic, not the aircraft: a model that brings dinner today, adapted for orbital logistics, could help feed astronauts in 2041.

2026

OBSERVED



Drones deliver food now

One delivery network passed 2 million commercial drone deliveries in January 2026, with a median flight of 3 minutes (Food On Demand).

TRANSFER: scheduling, tracking, exception handling

2030s

PROPOSED TEST



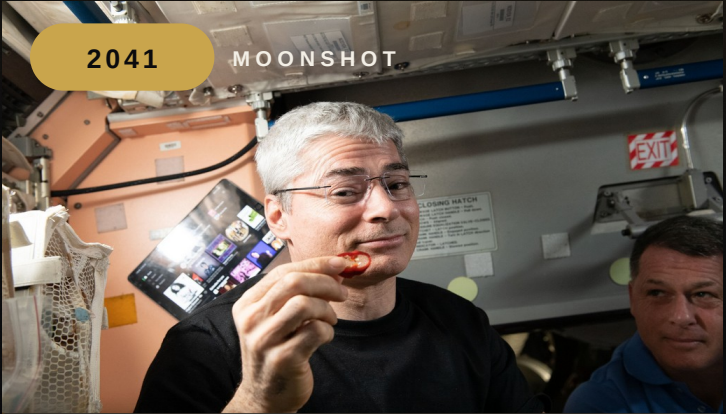
Autonomy reaches orbit

Robotic freighters already berth at the station using its robotic arm. Next: test AI scheduling with qualified space partners.

GATE: safety, communication delay, payload limits

2041

MOONSHOT



Feeding the crew in space

Each astronaut needs about 1.7 kg of food a day; fresh produce arrives every three months (NASA). Imagine cargo and orbital crops supporting a lunar outpost.

10x AMBITION: on-demand, not occasional, resupply

E,O&T test. Start with a logistics simulation and a partner review, not launch infrastructure. For clients in logistics and food, the last mile becomes the last orbit.

The new gold

TICK Gold is the treasure. TOCK Water, air, sand and plants are.

In 2041 value may sit in what keeps systems alive. Each card applies one of the 4 lenses of innovation and ends with an E,O&T opportunity.



Water

Training GPT-3 was estimated to evaporate about 700,000 litres of fresh water (Li et al.). The space station recycles 98% of its water (NASA).

LEVERAGE RESOURCES

Adapt closed-loop thinking to a client water audit.



Sand

Sand and gravel use is about 50 billion tonnes a year, 17 kg per person per day (UNEP). Desert sand is too round for concrete.

CHALLENGE ORTHODOXIES

Replace "abundant means disposable" with reuse by design.



Air

Lunar soil is 40 to 45% oxygen by weight (ESA), but it is chemically bound and not breathable until extracted.

HARNESS TRENDS

Track extraction tests, energy needs and readiness.

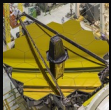


Plants

Lettuce and chile peppers grow on the station (NASA): crop research, not yet a complete food system.

UNDERSTAND NEEDS

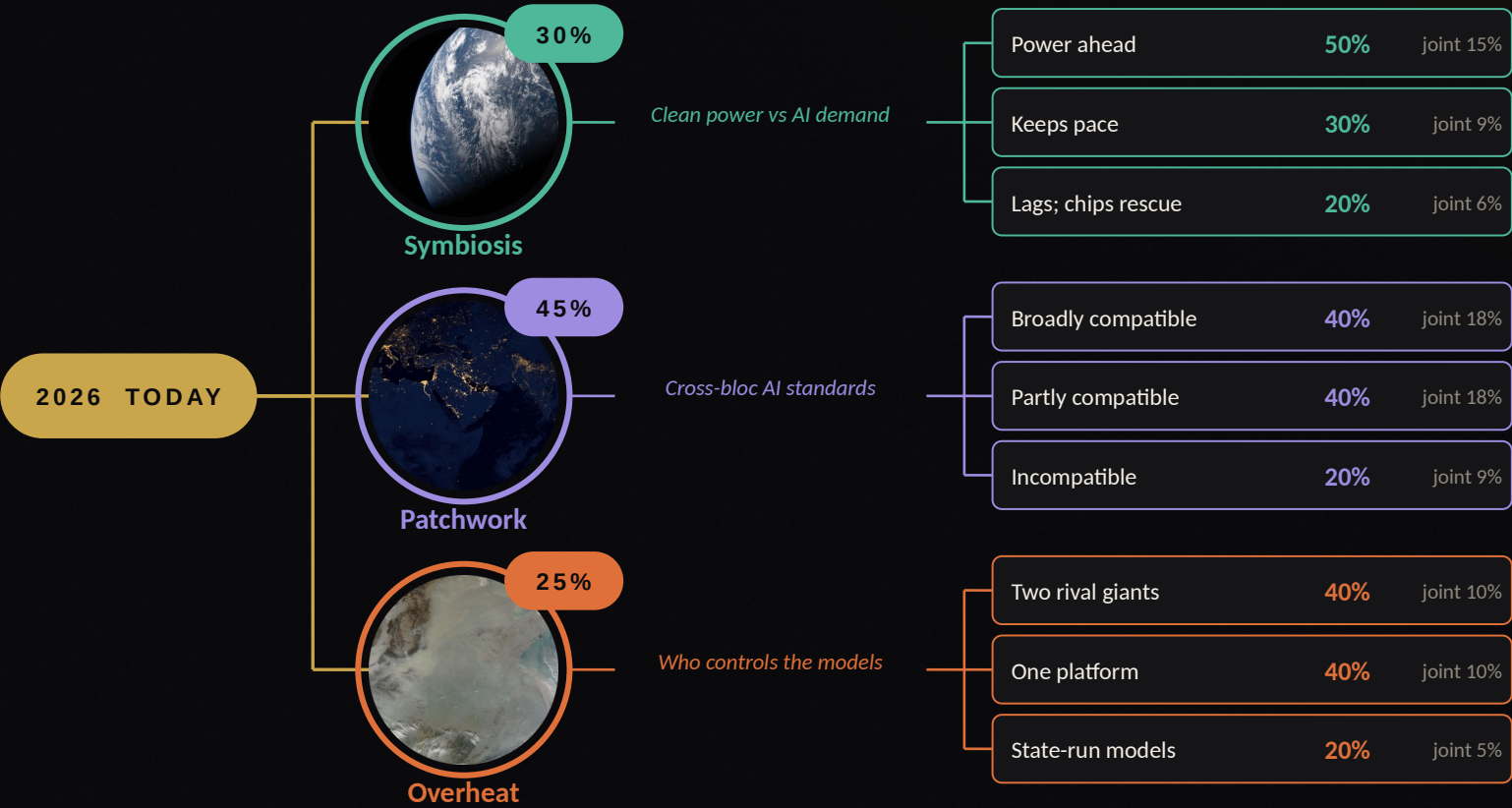
Design for nutrition, reliable supply and the crew's experience.



And gold? Still useful: it coats the mirrors of the Webb space telescope. But in 2041 it no longer measures wealth.

Which branch are we on?

Three worlds with planning weights, then one key driver per world split into three exclusive outcomes. Conditional odds sum to 100% per world, and the joint odds of all nine outcomes sum to 100%. Illustrative judgements, not calibrated forecasts.



How to read the tree

Level 1: worlds

Planning weights for the three worlds: 30%, 45%, 25%. They sum to 100%.

Level 2: one driver each

Three mutually exclusive outcomes per world, with conditional odds that sum to 100% within that world.

Joint odds

World x outcome. Worked example: Patchwork and partly compatible = 45% x 40% = 18%. All nine joint odds total 100%.

Why Patchwork leads

AI rules are already splitting by region; the clean-power race is not yet won.

Status

Illustrative judgements to support discussion, not calibrated forecasts.

A light only matters if it changes a decision

Each signpost has a record, a review trigger and a response. Triggers are proposed review rules measured against a pilot baseline, not external benchmarks.

SIGNPOST	WHAT WE RECORD	REVIEW TRIGGER	RESPONSE	POINTS TO
● Clean power for AI	Clean energy built per new data centre; energy used per completed task	Two monthly reviews worse than baseline	Switch to a smaller model or a manual route	Overheat
● Evidence quality	Share of outputs with a traceable source; material errors	Any unmitigated material error	Pause automation; require human approval	Symbiosis or Overheat
● Portability	Time and failures when moving one test case between providers	A second provider fails the same test	Add an adapter; avoid single-provider lock-in	Patchwork
● Orbital access	Debris-removal missions flown; launch delays	Removal still experimental by 2030	Plan ground fallbacks for satellite services	Overheat
● Human agency	Can users understand, challenge and reverse a result?	A high-impact decision has no appeal route	Redesign the handover before scaling	All three worlds

Cadence. The pilot lead records weekly, the client owner reviews monthly, and the team revisits the scenario assumptions every quarter.

Sell better decisions, not more reports

Three proposed offers, one per world, share one reusable asset: a Decision Passport that records why a client should act, who owns the decision and what happens if it fails.

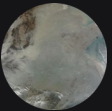


Future Design Lab

CLIENT Strategy and innovation leaders

Rehearse a new service in a digital twin before committing.

VALUE TEST Time to a testable decision



Verified Decisions and Continuity

CLIENT Operations and risk leaders

Check evidence, rehearse an outage and prove a fallback.

VALUE TEST Material errors; fallbacks completed



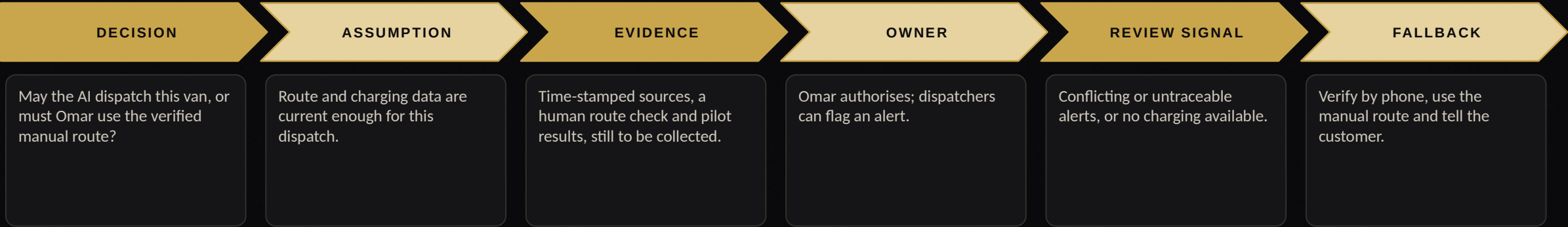
Interoperability Studio

CLIENT Product and technology leaders

Separate the core service from provider-specific adapters.

VALUE TEST Successful transfers; switching effort

DECISION PASSPORT | WORKED EXAMPLE: OMAR'S DISPATCH IN OVERHEAT (ILLUSTRATIVE REHEARSAL)



Proposed service concepts and an illustrative rehearsal, not existing products or results.



One causal story inside one world

The worlds branch, but a story inside a world must follow cause and effect. All eight Story Spine steps for Omar in Overheat, a fictional rehearsal.

01

ONCE UPON A TIME

Omar ran a depot whose customers depended on daily deliveries.

02

EVERY DAY

His team dispatched the vans on the AI's recommendation.

03

BUT ONE DAY

Two alerts conflicted, and the planned van could not charge.

04

BECAUSE OF THAT

Omar paused the recommendation before sending a driver.

05

BECAUSE OF THAT

He checked the records and called the depot to verify the route.

06

BECAUSE OF THAT

He used the manual fallback and logged why the decision changed.

07

UNTIL FINALLY

A verified alternative kept the delivery moving.

08

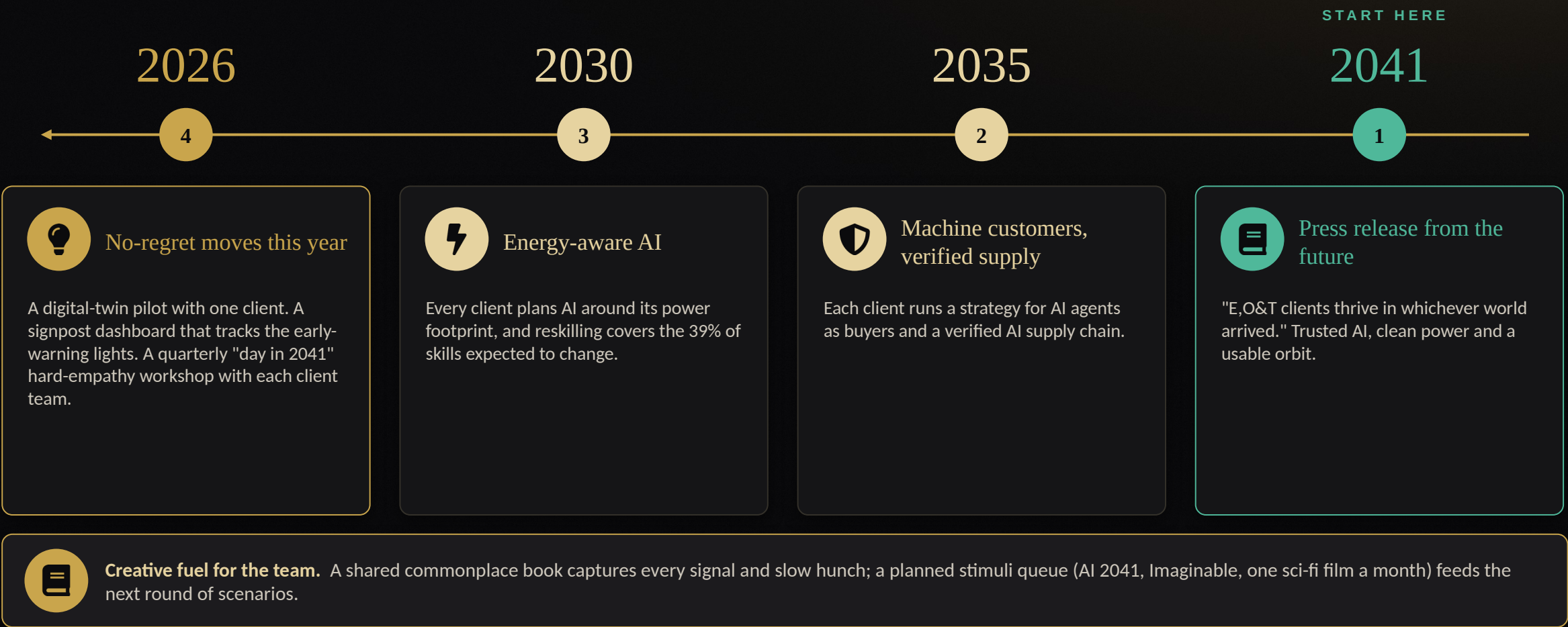
AND EVER SINCE THEN

The team reviews every exception with a Decision Passport.

Takeaway (5 Cs carry-out). The cure is not perfect AI. It is a process that keeps evidence, ownership and a fallback visible.

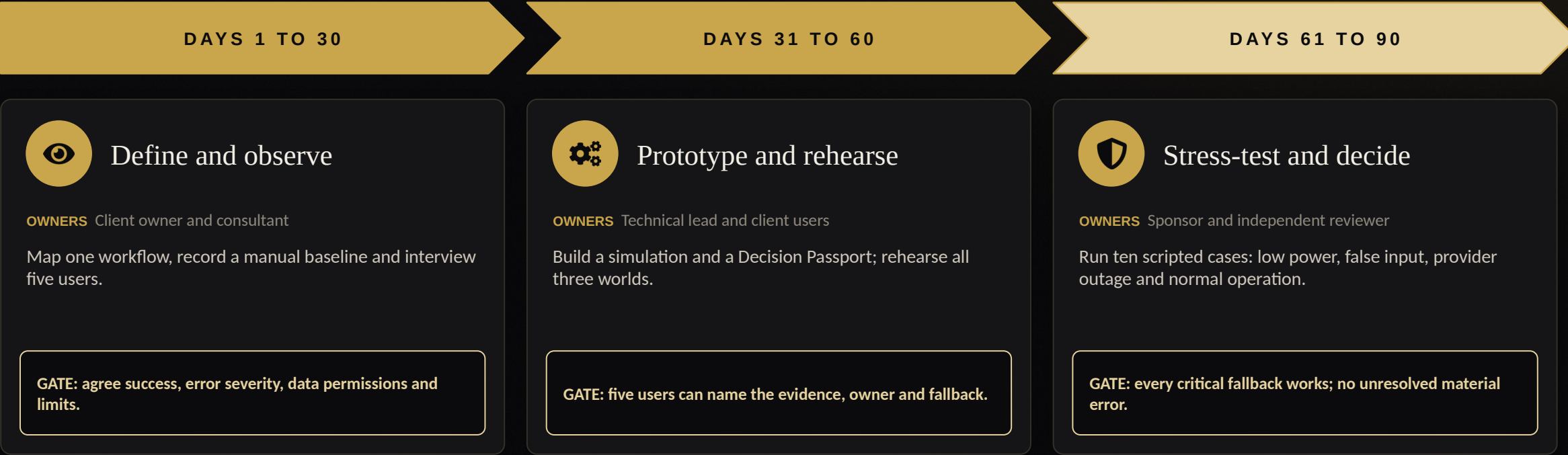
From 2041 back to Monday morning

How the worlds become action. We start at the end in 2041 and plan step by step, right to left, so every move made today is useful in any world.



Prove it is useful before scaling it

One client, one workflow, one limited rehearsal. All activities and gates below are proposed; no client commitment or benefit is claimed.



Continue, change or stop. Scale only when the pilot shows client-valued benefit and acceptable resource use.

Pilot limit. Five users and ten cases support learning, not proof of general reliability.

Rehearse, reflect, feed the next idea

A proposed monthly workshop that links the creative conditions, motivating feedback and future-thinking tools from this briefing.

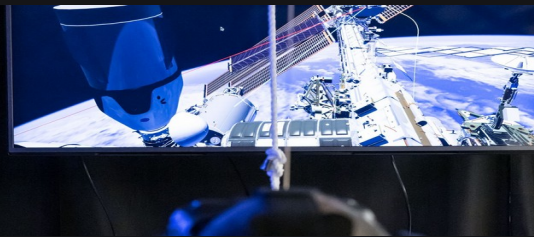


01 GENERATE | 10 MIN

A safe room for alternatives

PSYCHOLOGICAL SAFETY

Write ideas silently, then share. The leader invites disagreement; challenge the assumption, not the person.



02 REHEARSE | 20 MIN

Learn through the scenario

OCTALYSIS, USED ETHICALLY

Pick a world and a role. A shared progress bar tracks evidence, owner and fallback; retries are allowed after feedback.



03 REFLECT | 15 MIN

Keep the useful surprise

AI AND SCIENCE FICTION

Try one strange AI or sci-fi variation as an idea, not a fact. Check its assumptions and keep a testable version in the commonplace book.

Ethical design. Reward learning and feedback, not speed or a public rank. No penalties for failed ideas.

Monthly stimuli queue. One cross-industry source, one science fiction story and one observation walk; revisit slow hunches.

CLOSING

STORY SPINE Carry-out message

The Moon is closer than ever. *So is 2041.*

Three worlds, one discipline: make the assumptions visible, test the response, and keep a human decision owner in the loop.



Symbiosis

Build it



Patchwork

Shape it



Overheat

Insure against it

“ We do not predict the future. We prepare for all three, and we work every day for the first one.

Mohammad, Senior Futures Consultant, Em, Oli & Timmonds

NEXT Part 2, my personal reflection on the tools behind this briefing.



What I am taking with me

The tools that resonated, the change in my thinking, and why it matters in my professional life.

Three ideas from this course changed how I think about planning: possibility trees, working backwards, and counterfactual thinking. Together they gave me a structured language for something my work has always demanded: decisions about a future nobody can see.

In twenty years of network engineering and digital transformation, I have learned that a platform chosen today may still shape a business in ten years. Building the three worlds of 2041 for this assignment showed me a different way to approach that kind of decision. Drawing the possibility tree forced me to give each branch a probability, and the less likely branches raised questions a single forecast would never have asked. Working backwards from a 2041 press release, the way Amazon writes the press release before the product exists, made the first steps for 2026 obvious. Counterfactual thinking kept the exercise honest: writing Overheat next to Symbiosis stopped me from falling in love with the optimistic future.

What resonated most is that creativity here is not a talent but a structured skill, wild ideas plus strategy.



20 years

in networks and digital transformation, where decisions have long consequences

How I will put it into practice

A realistic timetable, a review loop and a personal routine that continues beyond this course.

My plan is concrete. From November, every new technical proposal I write will open with a one-page working backwards brief describing the result for the customer in three years, the assumptions behind it and the first test. Once a quarter, starting in January 2027, I will build a possibility tree for one roadmap decision, with three branches, nine sub-branches and probabilities, and review it three months later to see which branch reality followed. Before each major design review I will run one upward and one downward counterfactual with the team and record the risks it surfaces.

Personally, I live between Riyadh and Stockholm, and I will turn those flights into fuzzy focus time: no screens, just my stimuli queue and a commonplace book, so slow hunches have somewhere to land.

I will measure success simply: by June 2027, I will have built two possibility trees, completed the first three-month review, and recorded whether the findings changed a roadmap decision.



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Image credits

Slide 1: "A Breathtaking Earthset from Orion", Artemis II crew, 6 April 2026. NASA (art002e021278).

Slide 3: Night Earth observation from the ISS, Expedition 40. NASA (iss040e085126).

Slides 2, 4 and 14: "Home in the Distance", Artemis II. NASA (art002e020922).

Slide 5: "Earth in View", Artemis II (art002e023506); CADRE rovers concept, NASA/JPL-Caltech (PIA26296).

Slide 6: "Haze over eastern China", NASA Goddard (GSFC_20171208_Archive_e001269); ISS Expedition 35 (iss035e017673).

Slides 7, 10 and 17: "Black Marble: Africa, Europe, and the Middle East", NASA Goddard (GSFC_20171208_Archive_e001591).

Slides 5 to 7 and 16: NASA VR training (jsc2026e000427), Advanced Air Mobility concepts (AAM-Concept-2021, automation concept), Mission Control (jsc2026e019925), Pleiades supercomputer (ARC-2008-ACD08-0272-005).

Slide 8: NASA drone flight test (AFRC2026-0152-22); Cygnus in Canadarm2 (iss071e418235); Expedition 66 chile peppers (iss066e023179).

Slide 9: ISS water recovery (iss072e143163); sand dunes and irrigation in Saudi Arabia (PIA11069); Earth limb and Moon (iss051e039977); Veg-01 harvest (iss040e009126); JWST golden mirror (GSFC_20171208_Archive_e000368).

World thumbnails on slides 2, 4, 10, 12, 13 and 17 reuse the images of slides 5, 6 and 7.

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Use of AI



I acknowledge the use of Claude by Anthropic (<https://claude.ai>) and ChatGPT by OpenAI (<https://chatgpt.com>) to help generate and develop scenario ideas and variants, critique and develop wording, framework applications, scenario logic and probabilities, design the Decision Passport, pilot and workshop, find and check sources, and arrange and format the slides. The prompts used include "Propose solution directions for each company and the course concepts they would apply", "Draft three different future scenarios for 2041 with personas, built on the course tools", "Check every slide against the assignment requirements and rubric", "Merge the strongest ideas from my two drafts into one deck" and "Build the deck as a PowerPoint with my photo and a dark cinematic layout". The output was used as a starting point that I selected, reviewed, edited and verified; the choice of company and direction, the final scenarios, and the reflection based on my own experience and plans are mine, and I remain responsible for the submission.